AtlasSAT-16U

Orbital Verification, Coverage & Revisit Analysis

Document 2A-1 — Full Technical Report

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Document information			
Project	AtlasSAT-16U / Souss Water Observation CubeSat	**Study status**	Preliminary research demonstrator
Orbital model	GMAT R2026a, 550 km concept, near-SSO J2	**Time span**	20 Oct–19 Nov 2026 (assumed epoch)
Study region	Seven Chtouka-Ait Baha communes	**Agricultural ROI**	396.47 km^2 provisional mask
Evidence base	v0.2 / v0.3 / v0.4 + final review	**Relates to**	1A, 1B, 2A (one-pager)

Executive summary — The preliminary 16U Earth-observation concept targets agricultural water monitoring south of Agadir. A reproducible GMAT J2 orbit and independent geometric access calculation were reported and audited through v0.4. For a **30 km nominal cross-track swath**, the sampled agricultural target set does **not** satisfy the **72 h maximum optical revisit target**. Even allowing individually idealised $\pm 20^\circ$ cross-track roll, the longest complete daylight no-access interval is **312.219958 h (13.01 d)**, established for a sampled Sidi Bibi agricultural target. The updated research mask measures **396.47 km^2** within a **1,173.92 km^2** seven-commune study boundary, but crop mapping, real pointing feasibility and physical orbit/geolocation accuracy are not validated.

Status / permitted interpretation. This is a *simulation and geographic screening study*, not a flight qualification, an official agricultural land inventory, a demonstrated imaging schedule or a verified orbit determination. The displayed numerical precision must not be interpreted as physical location/time accuracy.

Evidence status: Saved source, numerical evidence and original figures were inspected directly in the Codex workspace. This authoring revision runs no propagation or access solver. The current v0.2 raw-output integrity exception is recorded in Section 8.4; v0.3/v0.4 coverage evidence is unaffected.

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1 Purpose, scope and evidence status

This document records the preliminary orbit, observation-access and revisit trade for a single AtlasSAT-16U spacecraft with a five-band VNIR pushbroom payload. The objective is to determine whether the concept can revisit agricultural targets within 72 h and to preserve traceability from assumptions to quantitative evidence. The geographic target is the Chtouka-Massa agricultural plain south of Agadir, not the entire administrative Souss-Massa region.

The numerical baseline (v0.2) and broad-rectangle coverage release (v0.3) remain frozen. Version v0.4 reuses the existing J2 ephemeris, adds independently sourced agricultural geometry, updates orientation post-processing, and performs two discrete initial-condition sensitivities. A separate final review resolves the provenance of the maximum daylight gap and the incomplete southern greenhouse dataset. No new propagation or access solver was executed in the preparation of *this LaTeX manuscript*. RevD3.4 integrates the approved coastline-corrected Technical Atlas with its refined locator, the approved monochrome globe and three-panel Sentinel-2 passage illustration, and original v0.4 scientific figures. The new maps and orbital illustration change presentation only; they introduce no land classification, propagation or revisit recalculation. Evidence keys E01–E18 identify bundled source excerpts in Section 10.2.

1.1 Four distinct evidence levels

Table 1: Evidence levels and permitted interpretation

Level	What this study establishes	What it does not establish
Numerical verification	Bounded agreement of independent implementations and numerical refinements	Orbit truth or a statistical flight-performance confidence level
Geometric reachability	A visible ellipsoidal target lies inside an ideal per-target angular envelope	A common pointing schedule or a complete acquired image strip
Daylight optical opportunity	Reachability plus a geometric target-Sun elevation gate	Cloud-free, quality-qualified or actually collected imagery
Physical/operational validation	Not demonstrated by this study	Qualified payload/ADCS, orbit knowledge, scheduling and delivered mission service

Evidence: E05, E10; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

1.2 Research question and decision rule

A mission-level requirement of at most $T_{\mathrm{req}} = 72\,\mathrm{h}$ is interpreted here as a maximum interval between optical access opportunities at each sampled agricultural target, including observed leading/trailing no-access spans. A sample failing the requirement disproves compliance for the sampled scenario. Satisfying every sample for a finite month would *not* establish all-season or continuous mission compliance.

1.3 Document family

Table 2: Document family and roles

ID	Artifact	Intended use
2A	One-page orbital analysis	Portfolio summary with principal evidence and limitations
2A-1	This detailed technical report	Assumptions, mathematical formulation, results and verification detail
2A-2	Analytical companion	Derivations, independent checks and comparison limits (E18)
V0.2–v0.4	Numerical / geographic evidence releases	GMAT files, source-code and result archives retained in Codex workspace

2 Mission inputs and requirement traceability

2.1 Declared parameters

Table 3: Mission inputs and assumption status

Parameter	Adopted value	Status
Platform / payload	16U; five-band VNIR pushbroom	Concept
Orbit initialisation	$a = 6{,}928.1363$ km, $e = 0.001$, $i = 97.5^\circ$ in MODEq	Assumed input
Initial RAAN / AOP / TA	6.463170753756° / 0° / 0°	Canonical inputs
Initial coordinate system	EarthMeanOfDate (Earth origin; MODEq)	Initial element axes
Epoch	2026-10-20, 12:00:00.000 UTC; 30 d propagation	Assumed, not launch date
Local descending-node time	10:30 mean solar time	Target / calibrated initially
Cross-track swath	30 km nominal; 20/40 km sensitivity	30 km requirement target
Target nadir GSD	5–10 m	Requirement; unvalidated
Along-track angular slit	From 10 m projected sample at 550 km	Provisional geometry only
Roll access limits	Nadir; ideal $\pm 10^\circ, \pm 20^\circ$	Sensitivity, not ADCS capability
Illumination	Target Sun elevation $\geq 0^\circ$ (also $\geq 10^\circ$)	Geometric daylight gate
Optical revisit	Maximum no-access interval ≤ 72 h	**Not met in studied cases**
Agricultural inventory	Seven-commune screened mask, 396.47 km^2	Provisional research layer

2.2 Traceability extract

Table 4: Requirements-to-method traceability

ID	Check / method	Observable output	Present verdict
MR-05	Daylight per-target exact and censored gaps	$G_{\max}$ and number of failures	**Fail** (sampled baseline)
OR-01	GMAT J2 propagation / two-body reference	Elements, periods, rate, convergence	Numerically supported
OBS-01	Pushbroom LOS/ellipsoid, roll envelope	Entry/exit, required roll	Geometric only
GEO-01	Mask lineage / area quadrature	Areas, class QC, target count	Research-only / incomplete

MR-05 follows the requirements-template identifier; OR-01, OBS-01 and GEO-01 are study traceability labels, not claims of previously baselined formal requirements.

3 Orbital model and verification

3.1 Propagation and model limits

The nominal GMAT R2026a spacecraft uses Earth gravity *JGM2 degree/order 2/0*, a Prince–Dormand 7(8) integrator, maximum internal step 30 s and tolerance 10^{-11}; the refined case uses 15 s and 10^{-12}. The two-body reference switches gravity to 0/0. The model does not activate drag, solar-radiation pressure or third-body perturbations. Physical mass and aerodynamic/radiation coefficients do not affect this force setup. The human-maintained GUI-first nominal script (E15) is canonical for the initial elements and matches v0.3. GMAT's saved AOP of 360° is equivalent to the canonical 0°; extra serialized decimal places are not a retuning. Native ephemerides use an Earth-centred EME2000 Cartesian output frame with kilometre and km/s units.

For reference, the standard central-plus-J_2 acceleration can be written using Cartesian coordinates aligned with the gravity symmetry axis (x, y, z), radius r, Earth gravitational parameter μ, equatorial radius R_E, and dimensionless coefficient J_2:

$$\ddot{\boldsymbol{r}} = -\frac{\mu}{r^3}\boldsymbol{r} + \frac{3J_2\mu R_E^2}{2r^5}\begin{bmatrix} x(5z^2/r^2 - 1) \\ y(5z^2/r^2 - 1) \\ z(5z^2/r^2 - 3) \end{bmatrix}. \tag{1}$$

This equation is an explanatory first-order representation; the implementation follows GMAT's configured harmonics and reference constants. Coefficient normalisation and force-model constants must be checked against the preserved GMAT setup for source-code parity.

3.2 Analytical reference and Sun-synchronicity

$$T_K = 2\pi\sqrt{\frac{a^3}{\mu}}, \qquad n = \sqrt{\frac{\mu}{a^3}}, \qquad \dot{\Omega}_{J_2} \simeq -\frac{3}{2} n J_2 \left(\frac{R_E}{a(1-e^2)}\right)^2 \cos i. \tag{2}$$

From the stated osculating input the independent audit reports $T_K = 95.649865790$ min and a first-order nodal rate $+0.973652342^\circ/\mathrm{d}$. The fitted seven-day EME2000 osculating RAAN rate is $+0.978448473^\circ/\mathrm{d}$, compared with a mean-solar advance of $+0.985647332^\circ/\mathrm{d}$. These are not exactly equal: the baseline is *near* SSO. A first-order inclination estimate of 97.592938° is a design starting point, *not* an executed retuning or proof of exact SSO.

2A-2, Chs. 1–4, Eqs. (2.6), (3.11), Tables 8–10 (E18) distinguishes the 95.649866 min two-body Keplerian period from the saved 95.578336 min J2 nodal period. First-order mean-element rates and fitted osculating RAAN also have different averaging/frame assumptions; neither difference measures propagation error.

Table 5: Orbital timing and nodal-rate evidence

Independently recomputed / fitted observable	Result	Interpretation
Seven-day fitted RAAN rate	$0.978448473^\circ/\mathrm{d}$	Near SSO
Thirty-day fitted RAAN rate	$0.978486827^\circ/\mathrm{d}$	Finite-interval mean
First descending local mean solar time	10:30	Residual 5.36×10^{-7} s
Seven-day sampled LTDN drift	-11.963 s	First to last node
Thirty-day sampled LTDN drift	-52.003 s	First to last node

3.3 Numerical cross-checks

The seven-day nominal/refined native OEM comparison at identical UTC epochs reports a maximum position separation of 0.0036204 m; the thirty-day comparison has the same reported maximum. The two-body universal-variable comparison reaches a maximum 0.001301824 m position difference. The corresponding maximum native velocity differences are 0.003973 mm/s (nominal/refined J2) and 0.001427 mm/s (two-body analytic reference). These are numerical consistency metrics, not observed positioning accuracy. The archived v0.2 validation contained 36 assertions covering finite states/counts/cadence, seven-day completion, altitude bounds, OEM/report agreement, conservation, analytic two-body agreement, J2/central nodal rates, integration refinement and node-time/root/geodetic checks; the v0.3 study added 25 core and four supplemental numerical/geometric assertions. Their dependencies and shared checks preclude interpreting a summed test count as independent physical validation.

Evidence: E01: `seven_day_native`, `thirty_day_native`, `twobody_universal`, `nodes7`, `nodes30`; archived validation inventory; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

Table 6: Selected analytical verification comparisons (2A-2, E18)

Quantity	Analytical / independent check	GMAT / Python reference	Interpretation / evidence status
J2 RAAN rate (deg/day)	0.973652	0.978448	F/R: first order versus fitted osculating rate.
Sidi Bibi, ideal $\pm 20^\circ$ interval (h)	312.222563	312.219958	F/R: saved period and observed clusters; N node counts; $+9.377$ s.
Mean rolling 72 h daylight coverage, ideal $\pm 20^\circ$	33.436055%	33.436055%	N/R: same saved events/weights; independent aggregation.

A: analytical relation; F: first-order approximation; N: independent numerical calculation; R: archived reference. These are calculation-provenance codes, separate from the four mission evidence levels in Table 1. Full values, residuals and qualifications: 2A-2, Table 33 and the supplementary CSV.

3.4 Preserved near-SSO trajectory illustration

Figure 1 shows the accepted monochrome globe, retaining the physical orbit-to-Earth scale and the Earth orientation at the selected snapshot. It is an inertial trajectory illustration against a snapshot Earth, not an Earth-fixed ground track or a new propagated orbit. The separate dated terrestrial passages are presented in Figure 5.

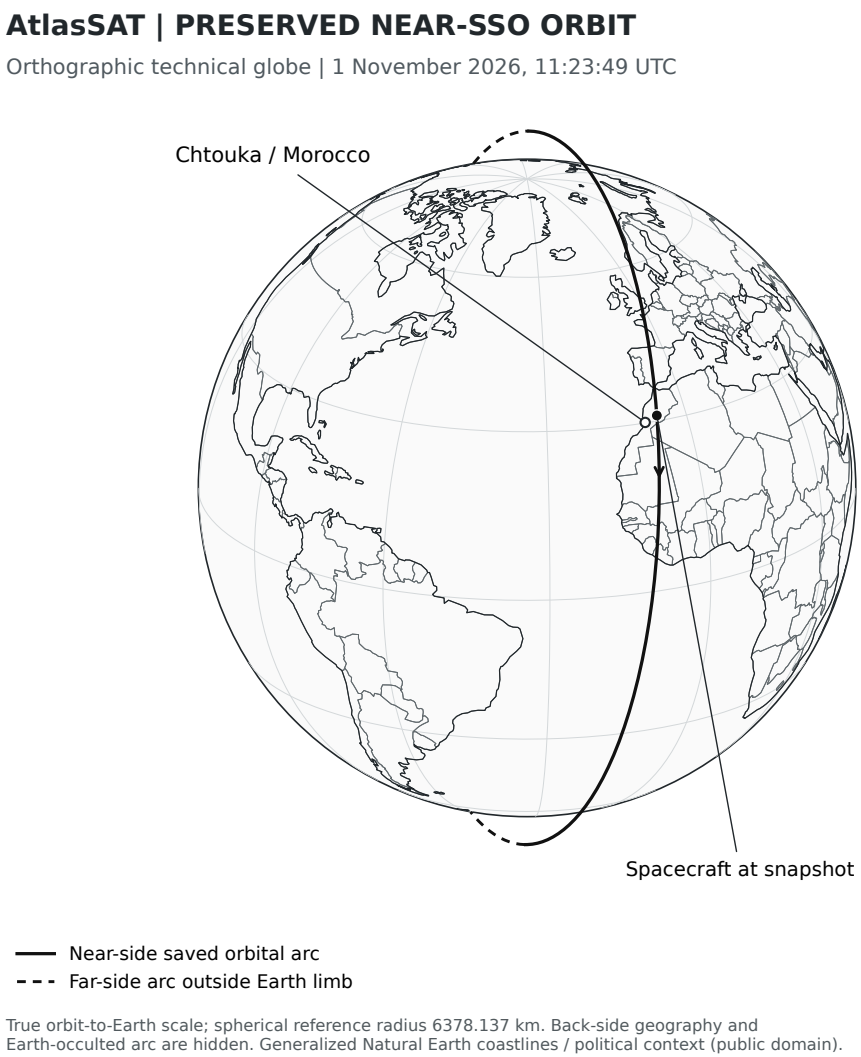

Figure 1: Approved monochrome orthographic globe. The preserved near-SSO arc uses a single snapshot orientation at 1 November 2026, 11:23:49 UTC; altitude is not exaggerated. Earth-occulted arc segments and back-side geographic boundaries are hidden; visible far-side segments outside the limb are dashed. The surface marker identifies Chtouka context, separate from the spacecraft marker. Predicted Earth orientation and a spherical display do not establish physical geolocation accuracy. E11/E14; Natural Earth public-domain geographic context.

4 Time, Earth orientation and coordinates

GMAT input spacecraft elements are expressed in EarthMeanOfDate/MODEq, whereas exported inertial OEM states are Earth-centred EME2000. Orbital-element values in these axes cannot be compared without their epoch-dependent transformations. For terrestrial geometry, the v0.4 post-processor applies a SOFA/ERFA mean-J2000 to GCRS/ICRF frame-bias correction and a SPICE J2000-to-ITRF93 time-dependent six-state transform, including the velocity term associated with rotation. UTC elapsed seconds are accumulated in TT then transformed to TDB/ET as required for SPICE evaluation.

The acquired Earth PCK `earth_000101_270116_261020.bpc` supplies Earth orientation for 2000–2027; its *last source input* was 20 October 2026 00:00 UTC, before the first simulated epoch. Subsequent dates in the study therefore use **predicted** orientation, not measured future UT1/EOP. The old and new orientation/frame procedures yield terrestrial position differences up to **103.00 m** (RMS 67.70 m) for identical inertial states; the frame-bias change by itself was ≤ 0.78 m at tested times. The matched opportunity-boundary shift across 28 targets is ≤ 2.872 ms, and a matched 1 km grid comparison changed mean 72 h area fraction by ≤ 0.0152 percentage point. Differences between model versions are not truth-error bounds.

Evidence: E06 kernel coverage, `combined_full_orbit_position`, `geographic_event_comparison` and paired-grid comparison; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

Interpretation control. The approximately microsecond event-root residuals and millimetre/centimetre state comparisons are numerical accuracy diagnostics. They do not establish precision ground geolocation, physical imaging instant, operational UT1, or validated orbit knowledge. Solar elevation is evaluated geometrically without terrain, atmospheric refraction or cloud screening.

5 Geographic region of interest and evidence quality

5.1 Outer boundary, mask and source classes

The seven selected community/OSM administrative commune polygons cover **1,173.917 km²**, comprising Aït Amira, Belfaâ, Biougra, Inchaden, Massa, Oued Essafa and Sidi Bibi. Names/codes were checked against the 2024 Moroccan HCP commune identities; this verifies identity, not the exact official vertices. Figure 2 shows the actual geographic definition. The seven-commune union is an explicit *study boundary*, not an official irrigation-service or aquifer perimeter. No reusable ORMVASM/ABH-SM/CRTS polygon or dated local crop inventory was obtained; the HCP check verifies identities only.

The screened primary agricultural research mask is **396.472 km²**: temporary-crop source support 261.79 km², mapped greenhouse support 129.28 km², and OSM orchard support 5.40 km². It combines 2021 ESA WorldCover and WorldCereal, a 2020 Global-PCG-10 greenhouse layer, OSM, and a documented October 2026 Sentinel-2 visual screen. The 456.37 km² candidate and 568.05 km² expanded alternatives reflect classification uncertainty rather than certified hectares cultivated in 2026.

Table 7: Study and agricultural areas by commune

Commune	Outer area, km²	Primary mask, km²
Aït Amira	176.18	90.15
Belfaâ	229.58	67.65
Biougra	10.63	2.56
Inchaden	143.04	57.63
Massa	167.88	6.47
Oued Essafa	271.28	126.39
Sidi Bibi	175.33	45.64

Rounded per-commune values do not necessarily sum to the higher-precision spatial union exactly. E02 gives exact class and subregion areas; E08 confirms administrative identities, not surveyed boundary accuracy.

5.2 Regional context and locator convention

The accepted Technical Atlas remains focused on Chtouka-Massa. Its relief transparency uses full-detail OSM land polygons on the unchanged 60 m grid; the terrain colours, multidirectional hillshade and 100 m contours derive from Copernicus DEM. Roads, rivers and verified locality nodes are OSM context; Oued Essafa is an italic commune-area label rather than an asserted settlement point. The small locator uses Natural Earth 1:10 million country geometry, following this publication's Morocco outline, with matching faint country context. These are traceable cartographic sources, not authoritative boundary or shoreline surveys. Coastline/generalization differences are not used to clip or relabel the mission mask. The national locator retains the regional study marker and the established publication outline; it does not set the mission-area denominator. No official provincial survey or agricultural perimeter is implied.

Evidence: E14 accepted-asset manifest and source notes; OSM contributors, ODbL 1.0; Natural Earth public domain; Copernicus DEM licence. E12 is retained as historical RevD2 context, not the active map source; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

5.3 Missing southern greenhouse coverage

The acquired Global-PCG-10 archive provides no corresponding source tile for the relevant study land south of approximately 30°N. Source-absent portions amount to **258.695 km²** (**22.04%**, rounded from 22.037%) of the outer study boundary and **27.201 km²** (6.861%) of the existing primary mask, affecting Belfaâ and Massa. The candidate and expanded masks have 38.203 and 56.974 km² respectively outside the source footprint. Those areas are *not* estimates of missing greenhouse acreage: unrepresented greenhouses could exist inside or outside the identified agricultural masks. The 2020 greenhouse raster and 2021 crop maps are not a validated 2026 crop census.

Evidence: E08 native source footprint and equal-area intersections; E02 source vintages and mask totals; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

Figure 2: Accepted Technical Atlas, using unchanged crops of its 600 dpi export: detailed OSM land/sea separation, Copernicus DSM relief and the refined Natural Earth locator. The seven-commune boundary and primary mask are unchanged, provisional research geometry. The dashed line marks the southern greenhouse-source limit. *The Morocco locator follows this publication's outline convention; geographic context only. WGS84 / UTM 29N display, original LAEA area measurement. OSM (ODbL); ESA/PCG (CC BY); Copernicus; Natural Earth (public domain). E02/E08/E10/E14.

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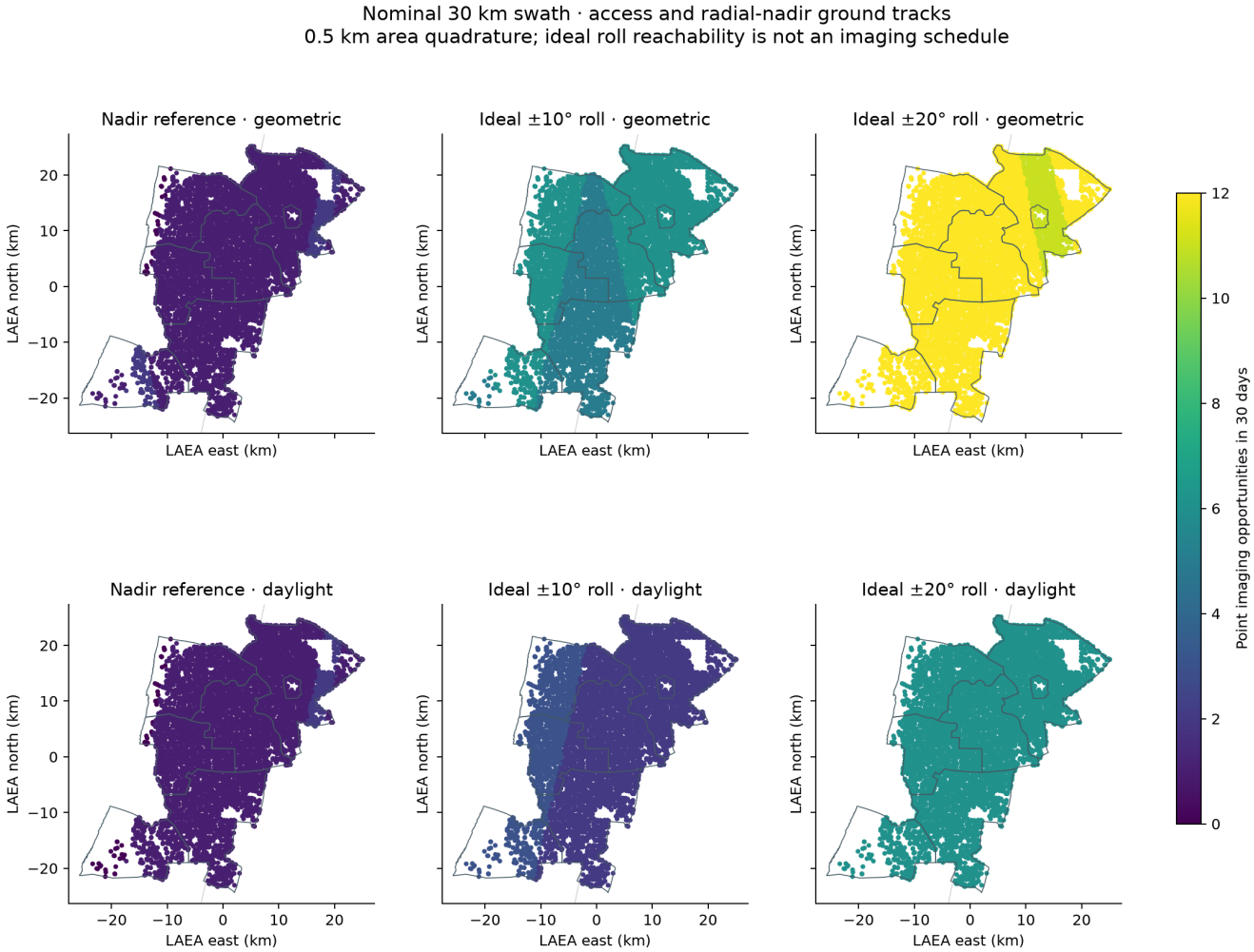


Figure 3: Original v0.4 access and ground-track figure, reproduced without data changes. Rows separate geometric reachability from daylight opportunities; columns separate nadir and ideal roll envelopes. The underlying map uses 500 m area quadrature, whereas principal coverage tables use 250 m. The radial-nadir track overlays are illustrative 30 s polylines; access events use continuous root refinement, not those plotted samples. Colour gives per-target opportunity count over the saved month, not acquired images. E03/E10; frozen `revisit_primary_500m.csv` and `groundtrack_30s.csv`.

6 Imager footprint and event modelling

6.1 Pushbroom geometry

With nominal altitude $h = 550$ km and nadir swath $W = 30$ km, a planar estimate for the full cross-track angular field of view is

$$\theta_{\text{cross}} = 2\tan^{-1}\left(\frac{W/2}{h}\right) = 3.124449834^\circ. \tag{3}$$

The provisional 10 m along-track surface sample yields a full along-track angular slit of $2\tan^{-1}[(0.010/2)/550] = 0.001041741^\circ$. Figure 3 illustrates the resulting observation corridors. The model treats the target and spacecraft as continuous geometric points; at the reported ground-relative speed the corresponding point residence windows are approximately 1.412–1.559 ms. This does *not* establish detector IFOV, commanded line rate, charge integration time, achieved GSD, multispectral alignment, modulation-transfer function or signal-to-noise performance.

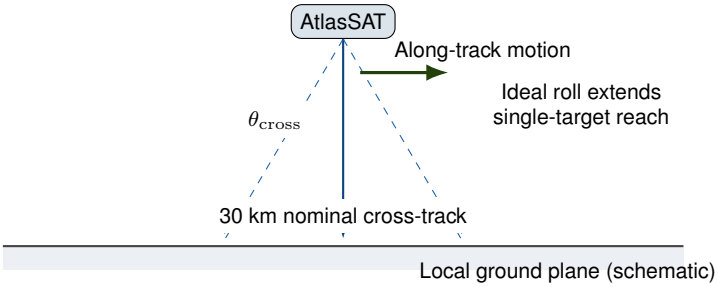


Figure 4: Conceptual viewing geometry, not to scale. The pushbroom's 10 m along-track slit is not resolvable in this drawing. A roll envelope describes independently reachable points, not executable slews or simultaneous coverage.

6.2 Coordinate basis and access predicate

In ITRF93, let target position be $\boldsymbol{q}$, spacecraft state $(\boldsymbol{r}, \boldsymbol{v})$ and line of sight $\boldsymbol{\ell} = \boldsymbol{q} - \boldsymbol{r}$. Radial nadir and ground-relative horizontal axes are

$$\boldsymbol{n} = -\frac{\boldsymbol{r}}{\|\boldsymbol{r}\|}, \quad \boldsymbol{u} = \frac{\boldsymbol{v} - (\boldsymbol{v}\cdot\boldsymbol{n})\boldsymbol{n}}{\|\boldsymbol{v} - (\boldsymbol{v}\cdot\boldsymbol{n})\boldsymbol{n}\|}, \quad \boldsymbol{c} = \boldsymbol{u}\times\boldsymbol{n}. \tag{4}$$

With $d = \boldsymbol{\ell}\cdot\boldsymbol{n}$, $x = \boldsymbol{\ell}\cdot\boldsymbol{c}$, $y = \boldsymbol{\ell}\cdot\boldsymbol{u}$, and full FOV half-angles $\theta_c = \theta_{\text{cross}}/2$, $\theta_a = \theta_{\text{along}}/2$, the implementation uses

$$\alpha = \operatorname{atan2}(x, d), \quad \rho = \operatorname{clip}(\alpha, -\rho_{\max}, +\rho_{\max}), \quad \beta = \operatorname{atan2}(y, d\cos\rho + x\sin\rho). \tag{5}$$

Access requires surface visibility $\boldsymbol{n}_g\cdot(\boldsymbol{r}-\boldsymbol{q}) > 0$, $|\alpha| \le \rho_{\max} + \theta_c$, and $|\beta| \le \theta_a$. Setting $\rho_{\max} = 0$ gives the nadir reference. The roll-dependent denominator is retained; this is not a sample-only or circular-footprint test. Each point's ideal roll is selected separately, so these predicates do not constitute an executable common pointing schedule.

Evidence: E10 METHOD/config; frozen v0.4 `src/coverage.py`; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

6.3 Continuous event detection

For irregular or widely spaced Cartesian ephemeris samples, the saved v0.4 pipeline uses position–velocity cubic Hermite interpolation and signed along-track fan-plane roots to find short target crossings between 30 s OEM records. Given endpoints $(\boldsymbol{r}_0, \boldsymbol{v}_0)$, $(\boldsymbol{r}_1, \boldsymbol{v}_1)$, $u = (t - t_0)/\Delta t$, the standard interpolant is

$$\boldsymbol{r}(u) = (2u^3 - 3u^2 + 1)\boldsymbol{r}_0 + (u^3 - 2u^2 + u)\Delta t\,\boldsymbol{v}_0 + (-2u^3 + 3u^2)\boldsymbol{r}_1 + (u^3 - u^2)\Delta t\,\boldsymbol{v}_1. \tag{6}$$

Evidence: E10 METHOD; angular FOV is a derived geometric estimate, not a verified instrument specification; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

The pipeline brackets fan-plane crossings, refines angular entry/exit roots and clips visibility/cross-track boundaries. A 30 s *sample-in-footprint* check cannot reliably discover millisecond point windows; discovery and refinement are verified separately against the saved independent checks (E05).

6.4 Daylight interpretation

If $\boldsymbol{s}$ is the unit target-to-Sun direction and $\boldsymbol{n}_g$ the outward ellipsoid normal at the target, geometric solar elevation is $E_\odot = \arcsin(\boldsymbol{n}_g \cdot \boldsymbol{s})$. Optical daylight opportunities require $E_\odot \geq 0°$, with a separate $10°$ sensitivity gate. Terrain shadow, cloud cover, seasonal crops and illumination-dependent image quality were not evaluated. Individual point windows do not represent the time necessary for full image-strip acquisition and spacecraft slew/settle.

The methodological companion is *Document 2A-2 — Analytical Derivations & Independent Verification*, v0.2.1 (49 pages, 27 October 2026; E18). It supplies step-by-step derivations, analytical predictions/approximations and independent consistency comparisons with archived GMAT/Python outputs, including assumptions, residuals and verification limits. It does not establish flight qualification or physical/operational mission validation.

2A-2, Chs. 6–8, Eqs. (6.1)–(6.3), (7.3)–(7.10), (8.1)–(8.5), Tables 26–28 (E18) derives pushbroom footprint, LOS/roll, illumination and event-boundary geometry, with worked local checks of millisecond point windows rather than an exhaustive ROI search. GMAT supplies native propagation and ephemeris export; Python performs agricultural optical-access, illumination and revisit analysis.

6.5 Illustrative passages from the preserved trajectory

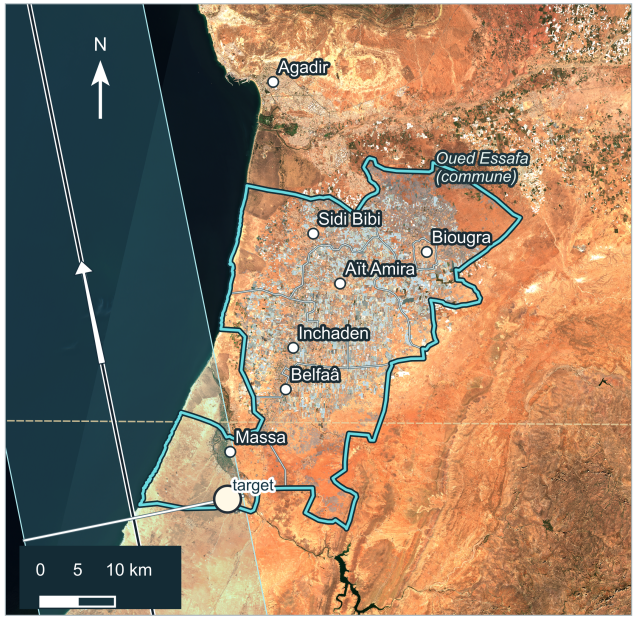
Figures 1 and 5 connect the preserved near-SSO orbit to the agricultural study. Event 1 is a saved nominal-swath nadir access to *Massa* on 24 October at approximately 22:51:42 UTC, with target Sun elevation $-62.32°$: geometric access at night. Event 2 is a corresponding *Oued Essafa* sample on 1 November at approximately 11:23:49 UTC, with Sun elevation $+43.13°$: a daylight opportunity, without a claim of usable acquired imagery. Event 3 is the radial central ground track inside the outer study boundary in two intervals within 16 November 11:24:31–38 UTC. It is a boundary-crossing diagnostic, not a full-ROI acquisition.

Exact selected point IDs, entry/exit times, source CSV lines and raw-file hashes are in E11. These are examples selected from existing results, not a new access or revisit calculation. The monochrome globe uses a single snapshot orientation at event 2; the regional panels use each date's terrestrial trajectory and separate illumination classification. Dates are not simultaneous states.

Evidence: E11 raw event selection and direct SPICE solar-sign checks; E10 radial-nadir convention. Numerical UTC precision is not physical event-time accuracy; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

24 OCTOBER 2026 | MASSA

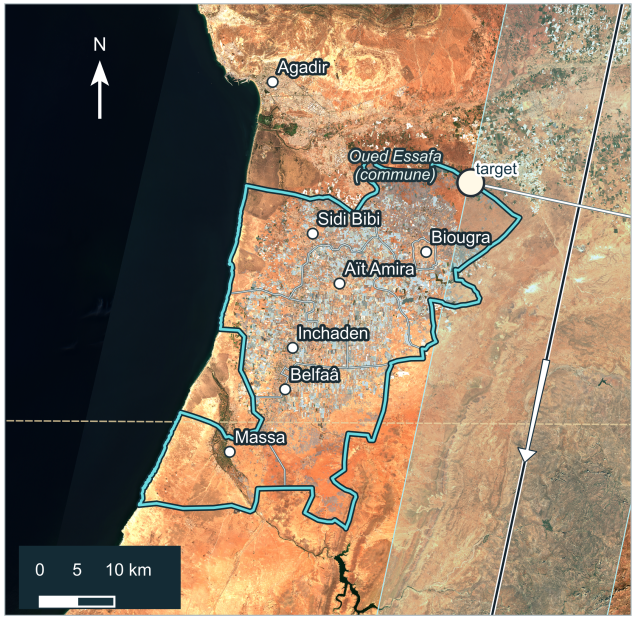
22:51:42 UTC | NIGHT



Saved agricultural target is reached by the nadir fan at night (Sun -62.32°). Ground passage and geometric reach do not provide daylight.

1 NOVEMBER 2026 | OUED ESSAFA

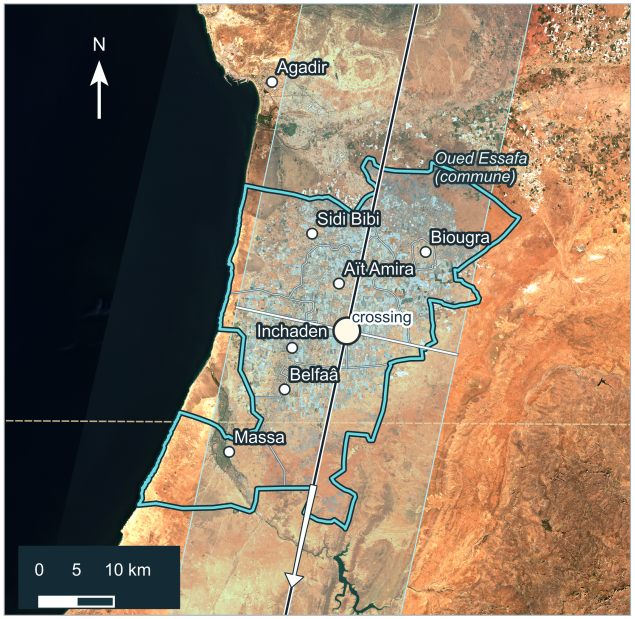
11:23:49 UTC | DAYLIGHT



Saved agricultural target is within the nadir fan in daylight (Sun +43.13°). This is an optical opportunity, not validated image acquisition.

16 NOVEMBER 2026 | CENTRAL TRACK

11:24:31-38 UTC | DAYLIT CROSSING



The central track enters the outer boundary in two separate intervals. Illustrated point is daylit (Sun +39.11°); full-ROI imaging is not implied.

HOW TO READ THE GEOMETRY

Seven-commune study boundary

Primary agricultural mask (provisional)

Ray-traced swept nadir fan

Radial-nadir ground track

Instantaneous cross-track fan

Swath is the ray-traced nadir fan swept over ±12 s around each illustrated epoch; not a simultaneous footprint or acquisition schedule.

Same basemap: real Sentinel-2 L2A, 12/20 Oct 2026; it does not depict event-time illumination. Provisional ROI; predicted Earth orientation.

Contains modified Copernicus Sentinel data (2026). © OSM contributors (ODbL); ESA / PCG (CC BY); Natural Earth. WGS84 / UTM29N.

Figure 5: Three separate epochs on the same Sentinel-2 background and map scale: Massa night access, Oued Essafa daylight opportunity, and a central-track boundary crossing. The shaded corridor is a ray-traced nadir fan swept around each event, not a simultaneous footprint or executable image strip. The imagery does not depict event-time illumination. Provisional ROI; predicted orientation; OSM (ODbL), ESA/PCG (CC BY), modified Copernicus Sentinel data and Natural Earth. E11/E14.

7 Coverage, revisit and requirement decision

7.1 Metrics and censoring

For target j with qualifying access intervals $[s_{j,k}, e_{j,k}]$, complete optical no-access gaps are $G_{j,k} = s_{j,k+1} - e_{j,k}$. Revisit midpoint differences can also be tabulated but should not be silently substituted for $G_{j,k}$. Left/right censored spans run from simulation start to first entry and last exit to simulation end; when no access exists the entire 720 h interval is an observed no-access span. These spans are lower bounds on unknown adjacent full intervals. A target with zero access has *undefined*, not zero, complete revisit statistics.

For a time window $[t, t+H]$, the area-weighted sampled coverage fraction is

$$C_H(t) = \frac{\sum_j A_j \, \mathbf{1}\{\exists k : \; m_{j,k} \in [t, t+H]\}}{\sum_j A_j}, \qquad m_{j,k} = \tfrac{1}{2}(s_{j,k} + e_{j,k}), \tag{7}$$

where A_j is each representative cell's *actual agricultural intersection area*. The reported rolling averages use starts every hour over fully contained 24/48/72 h windows, and measure at least one geometric opportunity per sampled agricultural area cell, not an acquired or delivered image.

7.2 Principal thirty-day result

Table 8: Principal daylight opportunity results

30 km / daylight	Nadir	Ideal $\pm 10°$	Ideal $\pm 20°$
Mean rolling 24 h agricultural coverage	3.56%	7.42%	20.66%
Mean rolling 48 h coverage	7.38%	14.82%	28.53%
Mean rolling 72 h coverage	**11.48%**	**22.76%**	**33.44%**
Maximum complete daylight gap	360.01 h	360.01 h	**312.22 h**
Maximum observed span, incl. censoring	720.00 h	360.01 h	312.22 h

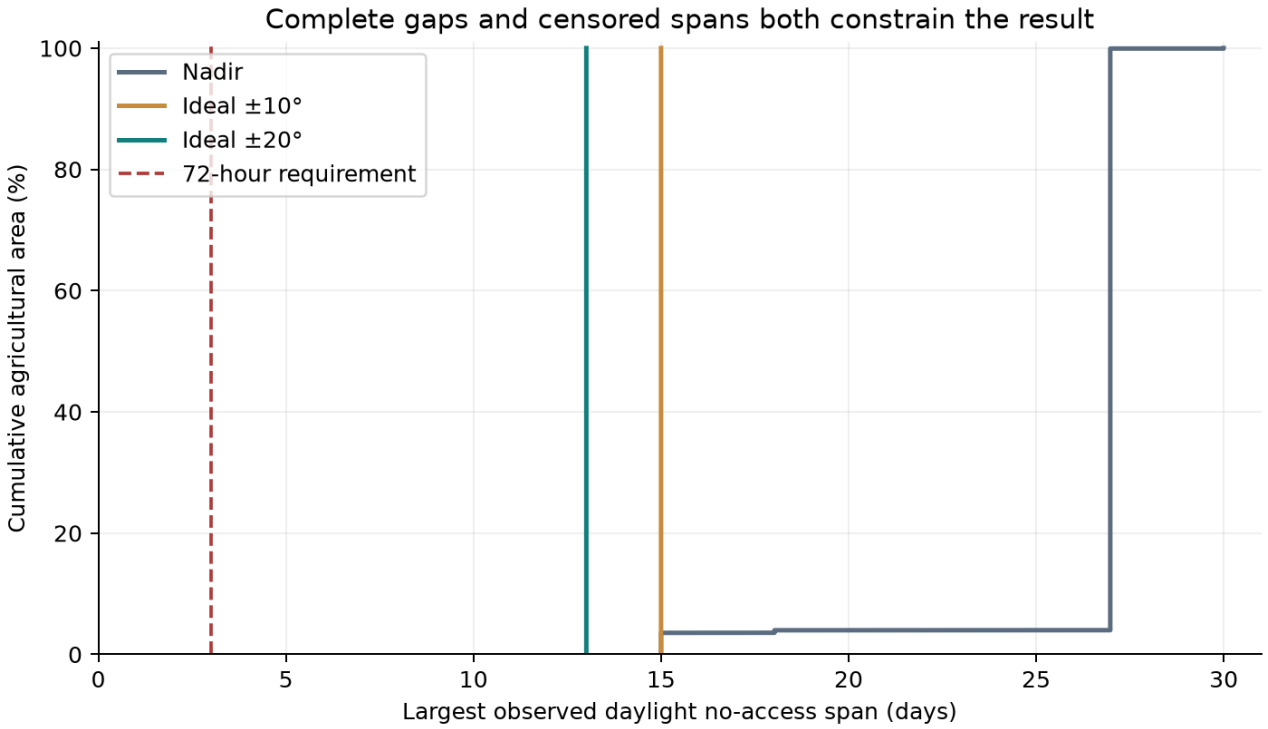


Figure 6: Original v0.4 gap-tail evidence, including censored spans. Horizontal position is each target's largest observed daylight no-access span; the cumulative agricultural area uses its GIS weight. The requirement is 3 days. Nadir includes targets unaccessed throughout the 30-day run, while even ideal $\pm 20°$ has gaps near 13 days. This is an opportunity test, not a schedule or image-quality result. E04: `exact_window_summary`; frozen `exact_window_gaps_primary_250m.csv`.

Figures 6 and 7 show complementary gap-tail and rolling-coverage evidence. As Figure 6 shows, the long tail remains decisive. The nominal sample contains **14,494** agricultural quadrature targets at a 250 m spatial

resolution. The release reports **14,494/14,494** with an observed gap above 72 h, including the idealised roll cases. The minimum rolling 72 h coverage is zero in all nominal-swath scenarios. A pooled $\pm 20^\circ$ complete-gap median near 23.90 h cannot establish a *maximum* three-day service: accesses cluster between longer outage intervals.

Evidence: E03 rolling-hourly means, failed-target counts and pooled complete revisits; E04 exact exit-to-entry gaps; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

7.3 Worst-case agricultural target — resolved provenance

Table 9: Worst complete daylight gap: agricultural provenance

Evidence field	Confirmed value
Agricultural target ID	`sidi-bibi_-10750_19500` (Sidi Bibi)
WGS84 target location	30.276400208635273°N, 9.61006239223738°W
Previous daylight window exit	2026-11-02 11:17:35.047317 UTC
Next daylight window entry	2026-11-15 11:30:46.896619 UTC
Exact complete no-access gap	**312.219958139439 h** (13.01 days)
Raw evidence	`results/access_primary_250m.csv`, lines 232405/232411

The supplementary audit reports that all 14,494 agricultural samples have a gap that rounds to 312.22 h (spread only 0.144 s), associated with the same 2–15 November access-pass pair. Figure 10 shows this pattern; it is *not* proof of an exactly repeating orbit or all-season recurrence. The historical 30.4°N / 9.6°W reference was explicitly *excluded* from the agricultural aggregate. Its previously similar 312.22 h value differs by about 0.015 s in the exact gap and agrees after rounding.

Evidence: E07: `primary_250m.exact_maximizers`, raw lines include the header; E08 verifies REF outside the study area; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

Optical revisit verdict: NOT MET for the sampled, provisional primary ROI at the inherited initial phase. Even ideal $\pm 20^\circ$ roll yields a complete no-access interval of 312.22 h, exceeding the 72 h design target by more than a factor of four. No inference that a different orbit, new payload, or scheduled ADCS would meet this requirement has been demonstrated.

7.4 Agricultural subregion comparison

Table 10: Subregion daylight coverage comparison

Mean rolling 72 h daylight coverage	Nadir	Ideal $\pm 10^\circ$	Ideal $\pm 20^\circ$
Aït Amira	11.09%	22.22%	33.44%
Belfaâ	11.09%	22.38%	33.44%
Biougra	11.09%	22.19%	33.44%
Inchaden	11.08%	24.22%	33.44%
Massa	10.95%	24.75%	33.44%
Oued Essafa	12.33%	22.19%	33.44%
Sidi Bibi	11.06%	23.92%	33.44%

These ratios apply to each commune's screened agricultural submask; they must not be interpreted as all parcels simultaneously imaged or to imply local greenhouse completeness.

Evidence: E03 subregion coverage; area fractions use each commune's own screened agricultural submask; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

8 Sensitivity studies and validation

8.1 Spatial convergence, classification and swath

The 250 m primary quadrature has 14,494 on-mask targets; the 500 m and 1,000 m alternatives have 4,254 and 1,262 respectively. For all tested 30 km/roll/illumination combinations, the maximum change in mean 24/48/72 h coverage from 500 to 250 m is reported as **0.00460 percentage point**. This supports convergence of *area-averaged* opportunity fractions, not proof of every field/edge access. The expanded 568.05 km^2 research mask shifts mean 72 h coverage by at most 0.1461 percentage point at the matched 1 km grid,

without removing sampled revisit failures. The unsimulated 456.37 km^2 candidate layer has a minor recorded polygon self-intersection; it cannot be presented as an analysis-ready geometry until separately repaired and versioned.

Figure 8 retains the saved sensitivity plot. The 20/30/40 km swath sensitivity shows only modest shifts in rolling mean over the same 1 km agricultural grid; the 20 km case is deliberately below the nominal design target. Enlarging angular FOV under fixed orbital conditions does not independently prove acceptable revisit or imager performance.

Evidence: E04 spatial convergence and expanded-mask sensitivity; E08 topology diagnostics; E10 findings and saved 1 km summaries; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

8.2 Orbit initial-condition sensitivity — not a pure phase trade

Two additional GMAT runs change input true anomaly to 90° and 180° at the same stated osculating semimajor axis, eccentricity and plane. For 30 km and ideal $\pm 20^\circ$ roll, 72 h daylight mean area coverage at 1 km was 33.44% (0°), 0% (90°), and 37.14% (180°). The 90° case yielded geometric opportunities but *only at night*; all 37,860 saved agricultural access instances had target solar elevation from approximately -64.85° to -61.74°. Independent checks reproduced the Sun elevation signs at tested events.

Figure 9 presents the saved spatial and initial-condition comparison. The final technical review found that **TA90 also changes subsequent mean orbital period**: mean descending-node period was 95.578336 min at the base phase, 95.967473 min at TA90, and 95.579186 min at TA180. Because J2 makes mean-element correction latitude dependent, holding *osculating* a fixed does *not* isolate a pure phase change at a common mean orbit. TA90's 15 nodal periods nearly equal 24 h (23.991868 h), helping maintain a repeating night-access corridor. These runs remain useful *initial-condition sensitivities*, not an optimised launch-phase study or a demonstrated route to optical compliance.

Evidence: E09 phase periods, direct Sun/frame checks and solar range. TA90's daytime branch misses the roll envelope; reachable ascending passes occur at night; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

8.3 Independent geometric checks and preservation

The v0.4 investigation reports an independent inertial Lagrange/direct-SPICE reference matching **570** opportunities across 31 geographically distributed targets and three roll scenarios, with maximum entry/exit discrepancy **1.197 μs**. The evidence also includes 468 forward ray/ellipsoid boundary and corner cases, 156 grazing discovery tests, four simulation-endpoint cases and 600-time state-interpolation comparisons (maximum position difference ~ 2.27 cm). Saved software QA reports 19 passing gates. All tests share some physical model inputs and do not establish operational pointing/image accuracy.

Evidence: E05 distributed events, forward-ray/grazing tests, interpolation and endpoint checks; E01 archived assertion inventory; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

8.4 Archive integrity: inherited RevD2 intake exception

The earlier final review recorded 378 frozen files unchanged at its own checkpoint. At the RevD2 authoring intake, two current v0.2 nominal output files already differed from that checkpoint: the OEM and state report stop at **26 October 2026, 06:15 UTC** (16,591 output records), rather than the archived seven-day end on 27 October at 12:00 UTC (20,161 records). The extant state-report records exactly match the archived prefix; matching OEM epochs differ only at export-roundoff level. This is a **pre-existing truncation discrepancy**, distinct from the historical creation-header-only mismatch.

No file was restored or rerun. Seven-day claims in this report cite the saved audit and original v0.2 ZIP, not a fresh verification of the truncated current files. The v0.3/v0.4 file inventories still match the prior final-review checkpoint; agricultural coverage and its negative verdict use those unchanged results. RevD3.5 preserves the files as found and compares all protected sources against its own intake snapshot; it does not repair the inherited exception.

Evidence: E13 inherited RevD2 integrity diagnostics; `qa/preservation_after.json` for this release; E01 historical seven-day evidence; field-level values and hashes in `QUANTITATIVE_TRACEABILITY.csv`.

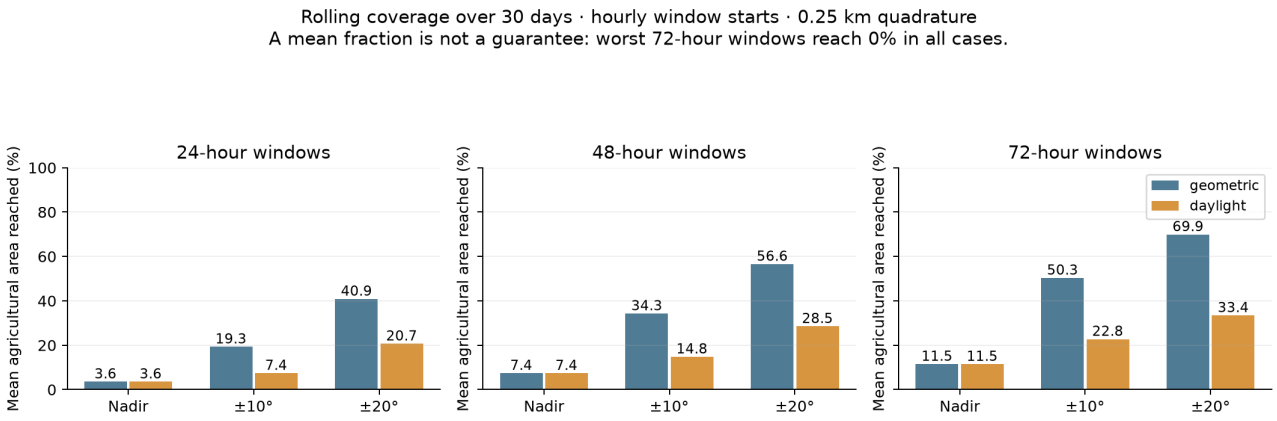


Figure 7: Original v0.4 rolling-coverage summary. The bars compare mean agricultural fractions across rolling 24/48/72 h window lengths and illumination classes; this is not an acquired-imagery timeline. Source values, scenario labels and the 250 m quadrature are unchanged. E03/E10; original figure and source sidecar retained.

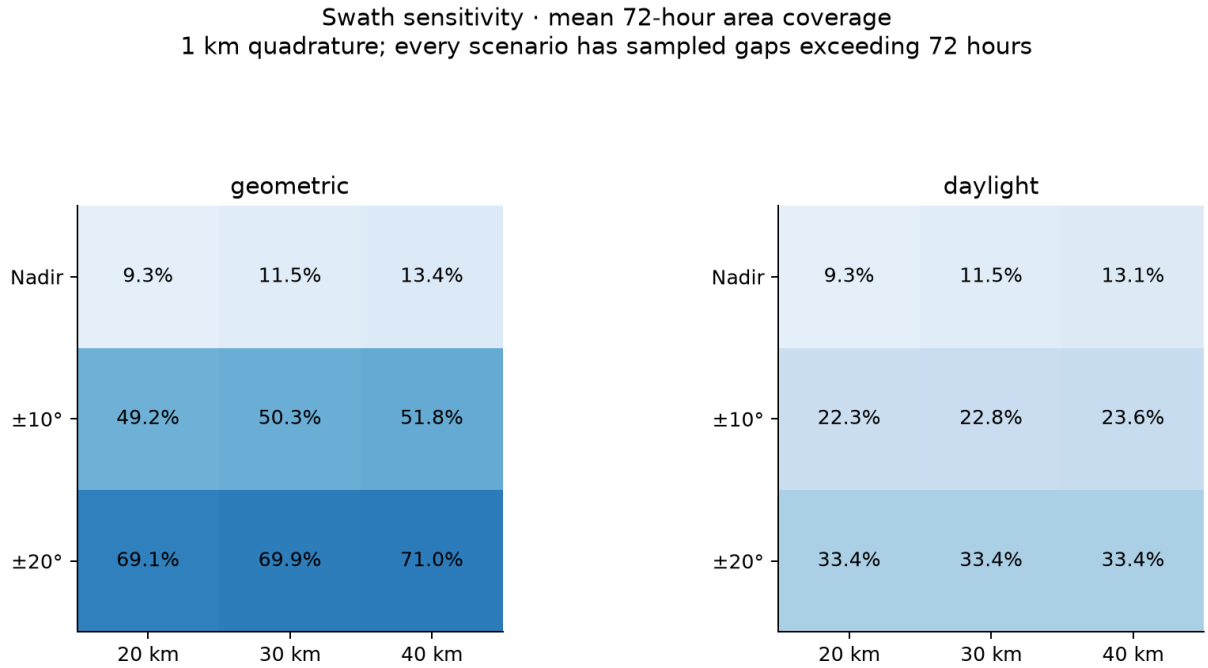


Figure 8: Original v0.4 nominal-swath sensitivity at the matched 1 km grid. The plotted swath and ideal-roll choices are geometric assumptions; none is a demonstrated payload or attitude capability. Changing swath does not by itself establish the maximum daylight-revisit requirement. E04/E10; original figure and source sidecar retained.

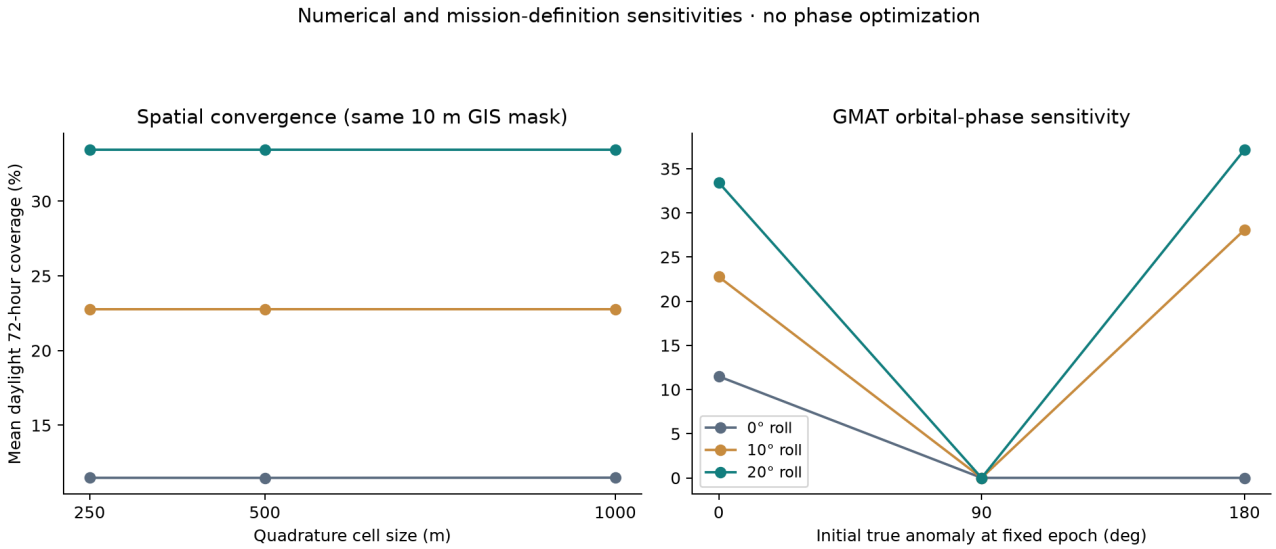


Figure 9: Original v0.4 spatial-convergence and initial-condition sensitivity figure. Area-mean convergence is numerical evidence, not field-level classification validation. The legacy phase axis denotes changed initial true anomaly at fixed osculating inputs; TA90 also changes the subsequent mean orbit, so it is not a pure phase-only comparison. The night-access explanation and independent solar checks are retained in the text. E04/E09/E10.

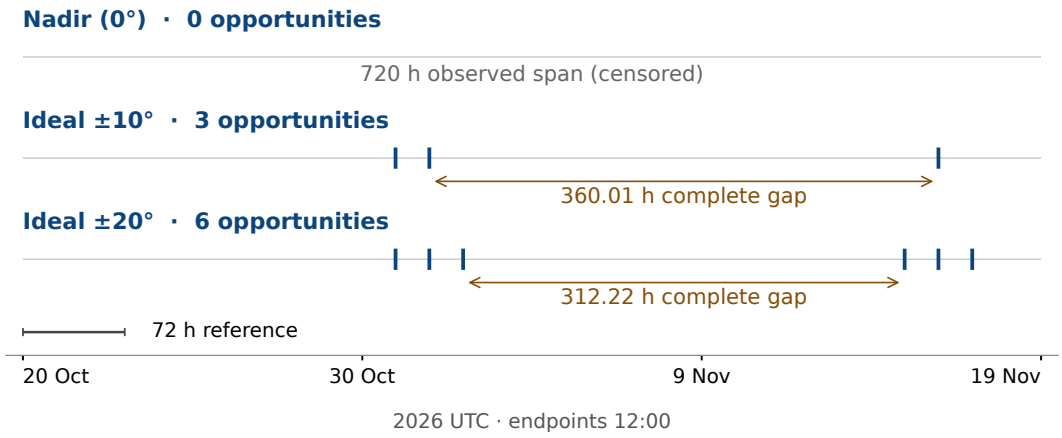


Figure 10: Saved daylight opportunities at `sidi-bibi_-10750_19500`, nominal 30 km swath, 20 October–19 November 2026 (12:00 UTC endpoints). The tracks contain 0, 3 and 6 opportunities. Nadir's 720 h no-access span is censored, not a complete revisit. Arrows mark complete exit-to-next-entry gaps for ideal roll; tick widths do not represent acquisition duration. E16, cross-checked against GUI-first three-target events.

9 Error budget, physical limitations and design implications

9.1 Evidence hierarchy and open uncertainties

Table 11: Evidence hierarchy and open uncertainties

Domain	Reported evidence	Limitation / open action
Integrator / two-body	mm-level cross-model numerical differences	Does not bound orbit truth
Footprint event boundaries	μs-level cross-implementation agreement	Shared geometry assumptions
Earth orientation / inertial alignment	Updated predicted EOP; 103 m model shift	No measured future orientation truth
Agricultural classification	Multisource 2020–2026 screening	No independent local crop truth; southern greenhouses missing
Orbit dynamics	30 d J2 near-SSO baseline	Drag, SRP, third bodies, dispersions omitted
Attitude/payload	Ideal roll + projected sample	No slew, settling, jitter, MTF, SNR, clouds
Coverage product	Representative-point quadrature	Not completed real image or full-strip schedule

9.2 Implications for a next engineering iteration

The sampled single-spacecraft baseline fails the daylight revisit target. A first-order check of the saved 95.5783355926631 min descending-node period gives about 15.06618 revolutions per day: slow ground-track drift clusters opportunities on a roughly 15-day scale (E17). This explains the scale, not exact access gaps. Next are phased 2–5 satellite constellation trades, followed by orbit-design, drag, solar-elevation and observation-scheduling studies; all remain unperformed. Attitude, power, storage, tasking and downlink need separate inputs and validation; see the supplementary roadmap. Local crop truth, terrestrial orientation error and real optical feasibility remain open.

2A-2, Ch. 5, §5.1, Eqs. (5.7)–(5.12), Tables 16–18 (E18) verifies 15 descending-node cycles within each selected ideal $\pm 20^{\circ}$ Sidi Bibi cluster, 226 between corresponding passes and 196 across the major gap. The ~ 15.111-day fractional beat differs from the ~ 15-day near return. The saved mean period and observed occupancy condition the timing model; independently checked node counts do not establish optical eligibility or event-search completeness.

2A-2, Chs. 9–10, Eqs. (9.1), (9.5)–(9.6), (10.2)–(10.3), Table 32 (E18) independently reconstructs gap arithmetic and reaggregates saved events/geographic weights: ideal $\pm 20^{\circ}$ mean rolling 72 h daylight coverage is 33.436055%, displayed as 33.44%. This verifies aggregation, not discovery of every optical event or actual acquired imagery; complete/censored spans remain distinct from ROI-wide area-weighted means.

Mission framing, ROI choices and review: Wassim Ait Oubahou. Codex assisted GMAT scripting, Python, validation, figures and documentation; the supplementary contribution note distinguishes these roles. Flight qualification is not claimed.

10 Reproducibility and references

10.1 Primary study evidence and traceable data products

RevD3.5 is a source-checked authoring layer on the saved numerical study. Selected JSON/CSV evidence, source hashes, figure sidecars and a field-level quantitative ledger are bundled with these documents. The large raw event and ephemeris archives remain in the frozen workspace releases. No new propagation or complete access recomputation was needed. The evidence locations are:

- `AtlasSAT_2A_v0.2/`: 7 d GMAT J2 and 0/0 propagation, calibration, OEMs, native logs and 36 checks.
- `AtlasSAT_2A_v0.3/`: 30 d propagation, preliminary rectangle, interpolation/geometry/solar validation, access CSVs and README.
- `AtlasSAT_2A_v0.4/`: seven-commune boundary, source data, 250/500/1000 m targets, agricultural masks, updated orientation, `FINDINGS.md`, `METHOD.md`, `DATA_SOURCES.md`, `TRACEABILITY.csv`, results and 19 software QA gates.
- `AtlasSAT_2A_v0.4_final_review/`: final provenance calculations for gap/greenhouse/TA90, scripts, evidence JSON/CSV and geometry figure.

Full solver reproduction requires the frozen sibling releases and their README/environment; this self-contained authoring build calls no GMAT or access solver. The separate GUI-first experiment provides an editable native propagation model; Python computes optical access, illumination and revisit. The canonical native GMAT script is included as E15 for local inspection and reproduction. Consider the inherited v0.2 integrity exception before historical reproduction.

10.2 Evidence register

The evidence register below uses keys E01–E18. Inspectable companion files are:

- `QUANTITATIVE_TRACEABILITY.csv`: values, units, saved fields and transformations.
- `evidence/source_manifest_RevD3_5.json`: hashes and original locations of copied sources.
- `ANALYTICAL_CROSS_REFERENCES.md`: exact 2A-2 locations, values and provenance.
- `MANIFEST.sha256`: authoring-deliverable hashes; original figure sidecars and the accepted-asset manifest are retained.

Table 12: Evidence register

Key	Evidence	Quantitative scope
E01	Saved independent orbital audit / archived checks	Period, precession, LTDN, integrator and two-body differences
E02	GIS area evidence / commune areas	Outer, primary, classes, candidate and expanded areas
E03	Primary 250 m summaries / subregions	Area-weighted rolling coverage, targets, midpoint revisits
E04	Saved synthesis	Exact gaps/censoring, grid convergence, geographic sensitivity
E05	Independent geometry, endpoints and QA	Bounded numerical validation counts and residuals
E06	Frame/orientation evidence	Kernel dates, terrestrial model differences and event shifts
E07	Final-review gap provenance	Sidi Bibi target, exact UTC windows, source lines and REF exclusion
E08	Final-review GIS / HCP / topology	Missing source footprint, identities and candidate caveat
E09	Final-review phase physics / solar evidence	Nodal periods, mean-motion qualification, night corridor
E10	Config, METHOD, data manifest and findings	Assumed mission inputs, equations, algorithms, data vintages/licences
E11	Selected raw passage events / saved OEM	Exact example targets, UTC windows, direct Sun signs, orbit/globe transform
E12	Historical RevD2 cartographic context	Retained provenance; superseded for active figures by E14
E13	Inherited archive-integrity diagnostics	RevD2 discovery of nominal v0.2 truncation; archived-results qualification
E14	Approved publication assets and source metadata	OSM coast, refined locator, DSM/Sentinel context, globe and ray-traced passage display
E15	Native initial-condition scripts and transfer checks	Full elements, epoch/axes, GUI AOP normalization
E16	Selected frozen daylight events / GUI-first cross-check	Sidi Bibi timeline, counts and complete/censored gaps
E17	Reproducible first-order arithmetic	Ground-track drift and recurrence-scale interpretation only
E18	Document 2A-2 v0.2.1 / 84-row comparison record	Derivations, independent checks, saved-event reaggregation and explicit limits

10.3 External technical references

1. NASA Goddard Space Flight Center, *General Mission Analysis Tool (GMAT)*: https://github.com/nasa/GMAT.
2. NAIF/JPL, *Frames Required Reading*: https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/req/frames.html.
3. NAIF/JPL, *SURFPT surface interception* and *illumination* documentation: https://naif.jpl.nasa.gov/pub/naif/toolkit_docs/C/cspice/surfpt_c.html.
4. ESA, *WorldCover 2021*, https://esa-worldcover.org/en/data-access; ESA *WorldCereal*, https://esa-worldcereal.org/en/products/global-maps.
5. OpenStreetMap contributors, *OSM administrative and agricultural land use* (ODbL 1.0); Niu et al., *Global-PCG-10* (CC BY 4.0); modified Copernicus Sentinel data (2026).
6. Natural Earth public-domain land, countries and populated places: https://www.naturalearthdata.com/. Active cartographic source records and licences are bundled as E14; E12 is historical context.
7. OSM land polygons, https://osmdata.openstreetmap.de/data/land-polygons.html; Natural Earth Admin 0 Countries, https://www.naturalearthdata.com/downloads/10m-cultural-vectors/10m-admin-0-countries/. Source snapshots and cartographic uncertainty are documented in E14.
8. Copernicus DEM GLO-30, https://registry.opendata.aws/copernicus-dem/; WorldDEM-30 attribution and licence accompany the authoring package. Sentinel-2 background imagery is geographic context, not a cloud-free imaging guarantee.
9. User-supplied Codex evidence, 20–23 October 2026: v0.3 independent audit; v0.4 findings and reproduction notes; and the separate final technical clarification addendum (Markdown).
10. Wassim Ait Oubahou, *Document 2A-2 — Analytical Derivations & Independent Verification*, v0.2.1, 27 October 2026, 49 pp. E18; complete comparison CSV, source traceability and validation/limitations records are bundled.

Final preliminary verdict. The v0.4 research-mask simulation is a transparent **negative design-feasibility result** for a 72 h *geometric daylight* revisit target under the given initial conditions. The computational evidence supports further design trade studies, not an operational performance or validated agricultural coverage guarantee. **Document 2A** gives the companion single-page portfolio summary.